

Note: real precedent research (Stages 1–2) is sourced; the scenario set (Stage 3) is explicitly forward-looking synthesis, not sourced fact.

Standardization in AI Compute, Inference/Training Infrastructure, and Messaging Business Logic: A Sanity Check, Precedent Analysis, and Scenario Set

TL;DR

- The upper layers of the stack (task format, context-appending, model routing, software stack) are standardizing fast and voluntarily — a competitively self-reinforcing equilibrium — while the physical/infrastructure and environmental layers standardize only under treaty-level or regulatory coercion, and remain competitively self-undermining absent an external forcing function. The OpenAI-compatible API, the Model Context Protocol (MCP), OCI/Kubernetes, and CUDA-vs-open-alternatives all confirm that de-facto-leader "race-to-standardize" dynamics govern the software layers.
- On the 6 scope items, the verdicts split: items 1, 2, 4 are standardized-emergent (de facto, no treaty); item 5 is standardized-with-known-origin (TIA-942, ISO/IEC 27001, Uptime Tiers, SOC 2, plus emerging export-control regimes); item 6 is standardized-with-known-origin in the EU/Germany (EED, EN 50600) but unstandardized elsewhere; item 3 (routing) is unstandardized-with-literature-on-tradeoffs. The single largest governance gap is exactly where the user's hypothesis pointed: routing/business-logic and the side-effects of API-call data reuse.
- The user's "global operating system for AI access" (impartial aggregate manager) is a **WEAK-to-moderate scenario**: the IAEA/nuclear analogy is strong for the *motivation* but breaks on verification (AI compute is not passively detectable like nuclear material), and the competitive logic cuts against voluntary adoption — labs defect from anything that dissolves brand attribution and customer lock-in unless a catastrophic incident or state coercion forces it, exactly as BGP security (MANRS/BGPsec) shows for a low-stakes analog.

Key Findings

Stage 1 — Per-item sanity check verdicts

1. **Task/prompt & dataset format → standardized-emergent.** The "OpenAI-compatible" Chat Completions API is the de facto interface; there is "no formally specified standard like HTTP or SQL... standardized by a governing body" (onprem.ai, Dec 2025), yet most providers offer OpenAI-compatible endpoints and middleware (LiteLLM) mirrors the format 1:1. Origin: organically emergent from incumbent dominance, no stated coordinating goal.
2. **Context/scaffolding/appending → standardized-emergent, unusually fast.** MCP, launched by Anthropic November 2024, was adopted by OpenAI (March 26, 2025, via Sam Altman on X), Google, Microsoft, AWS, and placed under Linux Foundation governance within ~1 year — "USB-C for AI." This is the clearest race-to-standardize case in the set. Origin now semi-formalized (Linux Foundation), but the coordination itself was emergent.
3. **Routing to a model instance → unstandardized-with-literature-on-tradeoffs.** No standard;

commercial aggregators (OpenRouter: 300–400+ models, 60+ providers; AWS Bedrock intelligent prompt routing) and academic benchmarks (RouterBench, RouteLLM, RouterEval) compete. Routing is deliberately proprietary/opaque — the exact locus of the user's hypothesis. (ZenML)

4. **Software stack (cloud inference/training) → standardized-emergent.** OCI + Kubernetes are de facto standards under the Linux Foundation/CNCF — Kubernetes holds a 92% share of the container orchestration tools market (SlashData/CNCF, with 5.6 million developers, 31% of all backend developers, using Kubernetes). The compute-language layer (CUDA) is proprietary and dominant — NVIDIA held 92% of the data center GPU market in 2024 (record ~\$115B data center revenue, up 142% YoY; AMD ~4%, per IoT Analytics) — with open alternatives (ROCm, SYCL/oneAPI, OpenCL, Vulkan) still trailing. This is a split verdict: standardized-emergent at the orchestration layer, partial race-to-fragment at the compute-language layer. (Grokipedia) (IoTAnalytics)
5. **Physical infrastructure → standardized-with-known-origin.** ANSI/TIA-942 (ANSI+TIA), Uptime Institute Tiers (I–IV; e.g., Tier I 99.671% to Tier IV 99.995% availability), ISO/IEC 27001:2022, SOC 2, ASHRAE 90.4, plus FedRAMP for government cloud; overlaid by export-control regimes (US BIS EAR, AI Diffusion Rule) as de facto access standardization. Multiple named bodies with explicit charters. (Data Center Asia)
6. **Datacenter environment/surroundings → standardized-with-known-origin in EU, sparse elsewhere.** EU Energy Efficiency Directive (2023/1791) Art. 12 + Delegated Reg. (EU) 2024/1364 mandate reporting for ≥500 kW facilities (24 data points → PUE/WUE/ERF/REF); Germany's Energieeffizienzgesetz sets binding PUE ≤1.2 for new facilities from July 2026. Elsewhere (US, most of Asia) largely voluntary. Noise/water/local-siting remains patchwork local ordinance — undecided/sparse-literature outside the EU. (Moduledge)

Stage 2 — Historical governance/standardization precedents

Internet governance bodies (primary priority). IETF (voluntary consensus standards, "rough consensus and running code," housed under ISOC), W3C, IEEE, and ICANN/IANA (unique identifiers; multistakeholder; IANA historically under US NTIA contract until the 2016 transition) form a multistakeholder model chosen over an ITU state-led model — a deliberate constitutional choice affirmed by the WSIS (2003/2005) process and the IGF (mandated by the 2005 Tunis Agenda). Regional Internet Registries (RIPE NCC, APNIC, ARIN) allocate address space. These are the closest structural template for AI's upper-layer standard bodies. (Public Knowledge) (Thegovlab)

MANRS / BGP routing security. Mutually Agreed Norms for Routing Security — voluntary, "honor system" after initial compliance checks. As of the Internet Society/GCA report "MANRS Participation Grows by 12.5% in 2024" (Feb 2025), "Nearly 1,200 networks and vendors are committed to making the global routing infrastructure more robust and secure" (NetOps participants grew from 904 to 1,012 in 2024, plus 141 CDN/Cloud, 31 IXPs, 6 vendors); the Global Cyber Alliance assumed the secretariat in 2024 under a five-year grant. Critically, BGPsec (the harder cryptographic standard) saw poor adoption (76% of surveyed operators had not even asked their vendor about it) because it offers no benefit unless near-universal and imposes a performance hit — the canonical race-to-fragment security-standard failure, directly analogous to the AI infrastructure-security layer. (Global Cyber Alliance) (Global Cyber Alliance)

IAEA/NPT dual-use governance. Treaty-based, near-universal, consent-based; verifies "outcomes rather than weapon designs" via material accountancy, on-site inspection, environmental sampling, satellite imagery; the Model Additional Protocol (INFCIRC/540, 1997) strengthened verification. The most-cited AI analogy — but the analogy breaks on physical detectability (nuclear material is radioactive and border-detectable; AI chips are not). (IAEA)

Comparator standardization cases from the 23-case catalog (justified inclusion):

- **LIBOR→SOFR (STRONG standardization precedent).** Benchmark-replacement; ARRC (convened by the Fed/NY Fed in 2014) chose SOFR; "hardwired" fallback language + the 2022 Adjustable Interest Rate Act (LIBOR Act) provided a legislative backstop; all 35 LIBOR settings permanently ceased 30 September 2024. Per the Bank of England (Oct 2024): "The transition away from LIBOR, once referenced in an estimated \$400 trillion of financial contracts, has made financial markets safer." Included because it is a *forced benchmark standardization* with a legal backstop — the model for a coerced AI standard. (Excluded as mere "general governance": it is specifically the replacement of one technical benchmark with another.) (Sofrrate + 6)
- **SWIFT / ISO 20022 (STRONG).** Community-chosen (2018) migration to a common messaging standard; coexistence ended 22 November 2025. Per a Swift spokesperson (Compliance Corylated, Nov 2025): "Adoption has been strong, with 80% of daily traffic now using the ISO 20022 format and more than 3.1 million payment messages exchanged daily across 210-plus sending countries and 220 plus receiving countries." G20/CPMI-endorsed. The cleanest "messaging business logic" standardization analog — voluntary but deadline-enforced by a member-owned cooperative. (Swift + 4)
- **Montreal Protocol (STRONG).** Substitute-technology-enabled; trade restrictions gave "enforcement teeth"; the Multilateral Fund financed developing-country transition; narrow set of chemicals + few manufacturers = tractable. The template for "standardize once a substitute exists and defection is penalized by trade exclusion." (ScienceInsights + 2)
- **Knight Capital / 2010 Flash Crash → circuit breakers + Market Access Rule (STRONG, post-hoc technical standardization).** The SEC imposed uniform single-stock circuit breakers (approved June 10, 2010), then Limit-Up-Limit-Down (2013) and the Market Access Rule (Rule 15c3-5, banning "naked" access, Nov 2010); explicitly because the disruption "was exacerbated by disparate trading rules and conventions across the exchanges" (SEC Chair Schapiro). The template for reactive, incident-driven AI infrastructure standardization. (Lexology + 3)

Submarine cable protection (comparator for infrastructure-protection treaty). The 1884 Convention for the Protection of Submarine Telegraph Cables (in force 1888; 36 states party as of 2013) criminalized cable damage and set a 1-nm ship stand-off; largely subsumed into UNCLOS but "hardly changed since 1884." Shows infrastructure-protection treaties are durable but slow and weak on high-seas enforcement (only the vessel's own flag state can act beyond the territorial sea) — a cautionary analog. (Wikipedia + 4)

Qualifying Q4 (mechanism mapping). Software-layer precedents (IETF/MCP/OCI) → scope items 1-2, 4; voluntary, no enforcement beyond network effects. Physical/environmental

precedents (TIA-942, EED) → items 5-6; binding via certification/audit or law. Dual-use precedents (IAEA, Montreal, export controls) → item 5 (access) and the routing/aggregation layer; binding via treaty + trade exclusion + (proposed) hardware verification.

Qualifying Q5 (hard-to-standardize client groups). Historically resistant: large enterprises negotiating bespoke terms; government/defense users (own networks, exemptions — the EED exempts defense/civil-protection facilities); high-frequency traders (pre-Market-Access-Rule "naked access"); spam/botnet operators and jurisdictions with weak enforcement; P2P and crypto-mining evading local rules. **AI analogs:** high-volume API resellers/aggregators; jailbreak/red-team communities; state-affiliated users; and gray-market GPU cloud resellers (the smuggling/diversion channel export controls already struggle with — a gray market growing "faster than the regulatory apparatus," per Chatham House). (Chatham House)

Qualifying Q6 (lowest-friction case + most-feasible AI analog). Lowest-friction precedent: **MCP / OpenAI-compatible API** (and historically TCP/IP, HTTP) — adopted in months, voluntarily, because a dominant incumbent's open, royalty-free format plus network effects made joining strictly better than defecting. Most analogous and therefore most-feasible-to-standardize-soon among the 6: **item 2 (context/appendix) and item 1 (task format)**, and secondarily **item 4 (orchestration software stack)**. Least feasible without treaty/coercion: **items 5 and 6 (physical + environmental)**, which historically required TIA/ISO certification regimes or EU law, and **item 3 (routing)**, where the competitive incentive is to stay opaque.

Qualifying Q7 (added): remote verification of AI compute. Because AI chips are not passively detectable like nuclear material, a body of 2023–2026 technical-governance literature proposes hardware-enabled mechanisms:

- Yonadav Shavit's "What does it take to catch a Chinchilla?" (arXiv 2303.11341, 2023): chip-firmware weight snapshots + training-run logging + supply-chain monitoring; estimates ~232,000 chip samples/year to verify a Chinchilla-280B-scale run. (arXiv) (ResearchGate)
- Mauricio Baker's "Nuclear Arms Control Verification and Lessons for AI Treaties" (arXiv 2304.04123, 2023), concluding verification challenges could be "reduced to levels that were successfully managed in nuclear arms control"; extended in Baker, Kulp, Marks, Brundage, Heim, "Verifying International Agreements on AI: Six Layers of Verification" (arXiv 2507.15916, 2025). (arXiv)
- CNAS/IAPS "Secure, Governable Chips" (Aarne, Fist, Withers, Jan 8, 2024), coining "on-chip governance" and proposing "Offline Licensing" (cryptographically signed compute allowances rendering smuggled chips useless). (arXiv)
- RAND's "Hardware-Enabled Governance Mechanisms" (Kulp, Gonzales, Smith, Heim, Puri, Vermeer, Winkelman, Jan 18, 2024), which notes tamper-proofing evidence "has been limited." (Scite)
- The ARIA-commissioned FlexHEG series (Petrie, Aarne, Ammann, Dalrymple, 2025; arXiv 2506.15093/03409/15100) proposing an auditable, open-source "Guarantee Processor" + tamper-proof "Secure Enclosure" acting as a substitute trusted third party. (arXiv)
- IAPS "Location Verification for AI Chips" (Brass & Aarne, 2024) using speed-of-light ping delay (est. \$25k/landmark/yr, 100–500 landmarks), now echoed in the pending Chip

Security Act (H.R. 3447). (laps)

- Real shipping tech: NVIDIA H100 Confidential Computing / TEEs with hardware root of trust, secure boot, remote attestation, and benchmarked <7% LLM-inference overhead.
- Sastry/Heim et al.'s "Computing Power and the Governance of AI" (arXiv 2402.08797, 2024) frames compute as uniquely governable because it is detectable, excludable, quantifiable, noting training compute rose "by a factor of 350 million" over thirteen years — while warning compute governance "is a double-edged sword." Caveat: nearly all *governance* uses are proposed, not deployed, and power-based workload detection has no published false-positive rate. (Governance) (Governance)

Details

The core competitive logic

The evaluative question — will competitors adopt standardization to avoid falling behind — resolves differently by layer:

- **Upper/software layers = race-to-standardize equilibrium.** MCP is the exemplar: OpenAI's decision to adopt a rival's protocol was "a remarkable strategic decision to join Anthropic instead of fighting" because a fragmented tool-integration layer was a pure cost with no defensible moat, and joining "neutralized any budding Anthropic advantage." The OpenAI-compatible API standardized because compatibility *expands* each entrant's addressable ecosystem. Kubernetes/OCI standardized under a neutral foundation "to avoid fragmentation and encourage innovation" (Docker's stated rationale for donating containerd). Here standardization is self-reinforcing: defection (a proprietary, incompatible interface) loses developers.
- **Compute-language layer = partial race-to-fragment.** CUDA is proprietary and dominant; NVIDIA's incentive is lock-in, and open alternatives (ROCm, SYCL, OpenCL) coordinate defensively but trail. Standardization here is contested precisely because the incumbent (holding 92% of data center GPUs) benefits from non-standardization.
- **Physical/environmental/access layers = coordination requiring external force.** TIA-942/ISO/Uptime standardized via certification markets (customers demand audited resilience) — a mild race-to-standardize driven by buyers, not peers. Environmental standardization (EED) required *law*. Access standardization (export controls) is *state-imposed* and actively fragmenting the ecosystem into "rival compute blocs" (National Interest) — the opposite of voluntary convergence. (The National Interest)
- **Routing/business-logic = self-undermining.** Opaque routing and undisclosed reuse of API-call data are competitive *advantages*; there is no network-effect reward for standardizing them and a real cost (loss of differentiation, exposure of data practices). This is why the user's hypothesis is correct that this is the least-governed layer.

Recommendations

Stage the analysis and any intervention by layer, matching mechanism to competitive logic.

1. Treat items 1, 2, 4 as already-solving-themselves. Do not expend regulatory capital

mandating prompt/context/orchestration formats; the market is converging (MCP, OpenAI-compatible API, OCI/K8s). Instead, *codify* the emergent standard through a neutral body (Linux Foundation is already doing this for MCP) to prevent late-stage capture. **Threshold to escalate:** if a dominant lab forks or closes MCP/its API to re-establish lock-in, move to consortium governance or antitrust attention.

2. **Prioritize item 3 (routing/business-logic) and the data-reuse side-effect as the true governance gap.** A light-touch disclosure standard — an "ISO 20022 for AI routing/telemetry": a structured, mandatory schema for *what was appended, where the query was routed, and whether the call feeds training* — would add the most marginal safety/transparency. Model adoption on SWIFT: member-owned cooperative + hard deadline. **Trigger threshold:** a single high-profile "silent routing to an unsafe/compromised model" incident (cf. the Hugging Face namespace-hijacking and malicious-model attacks — 352,000 unsafe-model issues, 335 traced to one "ClawHavoc" operation) should launch a Flash-Crash-style reactive standardization push. (TNW | Security)
3. **For items 5-6, rely on proven binding levers, not voluntary peer adoption.** Extend EED-style mandatory PUE/WUE reporting beyond the EU; use certification markets (TIA-942/ISO) for resilience; and treat export controls + hardware-enabled verification (FlexHEG, location verification, TEEs) as the access-layer analog to IAEA safeguards. **Threshold that changes the recommendation:** if remote verification (location ping, power attestation) demonstrates a low, *published* false-positive rate, a treaty-grade "IAEA for AI" becomes feasible; until then it stays diagnosis-only.
4. **Watch the hard-to-regulate client groups** (API resellers, gray-market GPU clouds, state-affiliated users, jailbreak communities) as leading indicators of standardization failure — as spam networks and HFT naked-access were for prior regimes.

Stage 3 — Scenario set (forward-looking synthesis; NOT sourced fact)

STRONG scenarios (primary analytical set)

S1. "MCP-for-Provenance": a mandatory routing-and-data-use disclosure schema

- *Scope items:* 2, 3 (and the data-reuse side-effect).
- *Description:* A member-owned consortium of labs + cloud providers (SWIFT/ISO-20022 model) defines a structured, machine-readable manifest that every API response must carry: which model instance served the call, what system prompt/scaffolding was appended, and a flag for whether the call's data enters training. Adoption is voluntary-with-deadline, backstopped by EU AI Act transparency provisions.
- *Strength:* STRONG — direct analogy to ISO 20022 and MCP; clear mechanism; plausible adoption path via existing consortia.
- *Evaluative answer:* Partially corroborates. Providers adopt *disclosure of routing/model* readily (a trust feature, like ISO 20022's richer data), but resist the *data-reuse flag* (self-undermining — it exposes a monetization practice). Net: race-to-standardize on the transparency envelope, race-to-fragment on the sensitive field. Likely requires a legislative backstop (LIBOR-Act style) for the contested field.
- *Projected outcome category:* B (persistent gap, then clear improvement once a triggering

incident + legislative backstop arrive).

- *Illustrative gesture (hypothetical)*: After a 2027 incident in which an enterprise discovers its confidential prompts were silently used to fine-tune a competitor-facing model, a coalition ships "AI-CBPR+" manifests; within 18 months 95% of API traffic carries provenance headers, but the "training-use" bit is only made mandatory by statute two years later.

S2. "Compute-Safeguards Regime": hardware-enabled verification as an IAEA-analog access standard

- *Scope items*: 5 (physical/access), plus training-format (1).
- *Description*: Chip vendors ship governance-capable silicon (TEEs + location verification + offline licensing per FlexHEG/CNAS proposals); an international body verifies that above-threshold training runs occur only on licensed, attested compute, mirroring IAEA material accountancy but via cryptographic attestation rather than physical inspection.
- *Strength*: STRONG on precedent (IAEA, Montreal trade-exclusion, export controls already de facto), moderate on mechanism maturity (verification tech is proposed, not deployed; tamper-proofing evidence "limited" per RAND).
- *Evaluative answer*: Corroborates *among states*, refutes *among firms* absent coercion. Chip vendors adopt only if (a) it unlocks otherwise-blocked sales (the Nvidia location-verification cost-recovery logic) or (b) mandated (Chip Security Act-style). This is Montreal-Protocol logic: standardize once a substitute/benefit exists and defection is trade-penalized.
- *Projected outcome category*: D (ongoing management, incremental improvement) — verification asymmetry and great-power politicization limit it, as they do the IAEA.
- *Illustrative gesture (hypothetical)*: By 2028 a "Compute Safeguards Agency" convened by a US-China working group verifies attested training logs from three jurisdictions; a diverted cluster is caught by a failed location-ping, but enforcement stalls when the host state disputes jurisdiction, echoing New START verification disputes.

S3. "Reactive Circuit-Breakers for Autonomous Agents"

- *Scope items*: 3, 4.
- *Description*: After an incident in which an autonomous agent (via MCP tool-calls) cascades a harmful action across services, regulators + an exchanges-analog (major clouds) impose uniform "kill-switch" and rate-limit standards on agentic execution, exactly as the SEC imposed uniform circuit breakers + the Market Access Rule after the Flash Crash.
- *Strength*: STRONG — precise analogy to a documented case; mechanism (mandatory pre-trade-style risk checks) already exists in finance.
- *Evaluative answer*: Corroborates. Post-incident, providers *want* the uniform standard because disparate rules amplified the harm and the liability; a race-to-standardize triggered by a shared shock.
- *Projected outcome category*: A (acute challenge, prompt effective resolution) if the incident is salient and bounded; B if diffuse.
- *Illustrative gesture (hypothetical)*: A runaway procurement agent places millions in erroneous orders across three SaaS platforms in minutes; within a quarter the major clouds

adopt a common "agent limit-up/limit-down" API that pauses tool execution on anomalous velocity.

The document's own scenario (developed independently)

S4. "Global Operating System for AI Access" (impartial aggregate manager/distributor)

- *Scope items*: 3 (routing), 2 (appending), 5 (access/monitoring), touching 1.
- *Description*: Competing labs/nations establish an impartial aggregator that (a) obfuscates from users which lab/subnetwork serves their query (frustrating subnetwork-hijack attacks like the Hugging Face namespace/model-poisoning cases), (b) preserves per-session user tracking for intent-monitoring, (c) prevents single-lab lock-in, and (d) guarantees labs retain their customer base regardless of which lab's model carries a latent vulnerability. Grounded in IAEA dual-use governance and BGP/MANRS routing-security precedents.
- *Strength*: WEAK-to-moderate. The *motivation* (mutual-risk cooperation after an infrastructure attack; US-China Schelling-point coordination) is well-grounded; the *mechanism* is speculative and internally in tension.
- *Evaluative answer*: **Largely refutes** voluntary adoption. Requirement (c) "no lock-in" and requirement (d) "retain customer base regardless" are in direct competitive tension — an aggregator that dissolves brand attribution destroys the differentiation labs compete on, so each lab's dominant strategy is to defect (offer a superior direct API), exactly as BGPsec failed because unilateral adoption gave benefit only if universal and imposed a cost. It becomes adoptable only under (i) a catastrophic, attributable incident that makes fragmented access an existential shared liability (Montreal/Flash-Crash logic), or (ii) state coercion (export-control/IAEA logic). The obfuscation-plus-monitoring design also mirrors the confidential-routing vs. lawful-intercept tension.
- *Projected outcome category*: C (diagnosis only) in the near term — much proposed, little adopted — escalating to A/B only conditional on a forcing incident.
- *Illustrative gesture (hypothetical)*: Following a 2028 automated attack that hijacks a popular open model to exfiltrate data at scale, a "Geneva-for-AI" summit stands up a neutral routing fabric; two frontier labs join for reputational cover but quietly keep their premium direct endpoints, and the fabric handles mostly commodity traffic — a partial MANRS-like honor-system equilibrium.

WEAK / low-substance scenarios (thin-literature; lower analytical weight)

S5. "Open Compute-Language Mandate" (item 4). A public-sector procurement rule (FedRAMP-adjacent) requires portability across CUDA/ROCm/SYCL for government AI workloads, forcing an OCI-like open abstraction. *Evaluative*: refutes voluntary adoption (incumbent lock-in with 92% GPU share), corroborates only under procurement coercion. *Projected*: D. *Gesture*: A defense procurement clause requires SYCL portability; vendors comply narrowly for government tenders while consumer stacks stay CUDA-only. **[LOW-SUBSTANCE — thin evidence of demand-side push.]**

S6. "Global Datacenter Environmental Passport" (item 6). EED-style mandatory PUE/WUE/noise/water reporting harmonized globally via ISO/IEC + a public registry. *Evaluative*:

corroborates only where law compels (EU/Germany today); elsewhere self-undermining (reporting exposes cost/impact). *Projected*: *B* in regulated regions, *E* elsewhere. *Gesture*: An EU-style label spreads to two Asian jurisdictions after a water-scarcity siting backlash. [LOW-SUBSTANCE outside EU.]

S7. "MANRS-for-AI-Routing" (items 3, 5). A voluntary "Mutually Agreed Norms for Model Serving" — signatories commit to filtering poisoned models, attesting model provenance, and not serving hijacked subnetworks. *Evaluative*: corroborates weakly — honor-system, value-proposition-limited like MANRS; big players join for reputation. *Projected*: *D*. *Gesture*: 200 providers sign; measurement shows signatories filter malicious models more tightly, non-signatories lag. [LOW-SUBSTANCE — analogizes to a partially successful, non-binding regime.]

S8. "Dataset-Format Standard for Fine-Tuning" (item 1). A common schema (JSONL + provenance/consent metadata) for finetuning datasets, consortium-driven. *Evaluative*: corroborates — low-friction, TCP/IP-like, expands interoperability. *Projected*: *A/B*. *Gesture*: A Hugging Face-led schema becomes default; provenance fields lag adoption. [LOW-SUBSTANCE — plausible but sparse literature.]

S9. "Switch-Off Treaty" (item 5). A treaty-grade agreement (submarine-cable-convention analog) criminalizing unauthorized interference with, and guaranteeing state switch-off privileges over, frontier training clusters. *Evaluative*: refutes near-term voluntary adoption; treaty-dependent and slow (the cable convention has "hardly changed since 1884"). *Projected*: *C*. *Gesture*: A convention is drafted but, like high-seas cable enforcement, lacks a mechanism against non-flag actors. [LOW-SUBSTANCE — highly speculative.] The Conversation

Caveats

- **Real vs. hypothetical.** Stages 1–2 are sourced; Stage 3 (S1–S9) is explicitly synthetic — the outcome categories are *projected*, and the illustrative vignettes are invented, not documented.
- **Speculative-source hygiene.** Several verification mechanisms (FlexHEG, offline licensing, location verification, "IAEA for AI") are *proposals* using conditional language, not deployed systems; RAND flags tamper-proofing evidence as "limited," and no published study quantifies false-positive rates for power-based training detection. The Chip Security Act (H.R. 3447) is pending legislation. NVIDIA H100 TEEs are real and shipping but were built for commercial confidentiality, not governance.
- **Recency/AI-dated sources.** Some 2026-dated figures (adoption shares, forthcoming EU packages, market shares) come from vendor/industry sources and should be treated as directional. Export-control tier details shift frequently. The Swift "80% of daily traffic" figure (spokesperson, Nov 2025) and the "97% of payment instructions" figure (Swift press release the day after coexistence ended) differ in scope (steady-state daily traffic vs. instructions observed the day after cutover) — I report both rather than reconcile.
- **Conflicting framings.** Whether export controls constitute "standardization" is contested — they de facto standardize access while actively *fragmenting* the technical ecosystem into rival blocs; I treat them as access-layer standardization with a fragmentation side-effect.

